# Applicability theorem: closes Study 18 OPEN item

## Applicability of R(u) to density perturbations

### Linear regime ($\delta \ll 1$)

$\delta(x, t)$ is a fluctuation of the same field as the background
No system/spectator split $\Rightarrow$ axiom (A1) fails
$\Rightarrow$ R(u) does NOT modify linear growth
$\Rightarrow$ $\sigma_8^{ESD} = \sigma_8^{\Lambda CDM}$

### Nonlinear regime ($\delta \gtrsim 1$, virialized halo)

Halo is a localized subsystem against a separated background
Well-defined g, all three axioms hold
$\Rightarrow$ R(u) applies (Studies 09-16: RAR, SPARC, ...)

### Naive alternative (ruled out)

If (A1) held linearly: u(8 Mpc/h) = 7.4e-3, R(u) = 18.7
$\sigma_8$ would be boosted by 4.4x — incompatible with all data